SPE-229750-MS

Monitoring CO_2 Injection using DAS VSP Within a Carbonate Saline Aquifer. A Quantitative Approach Onshore Abu Dhabi

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

A CO_2 sequestration well was drilled in a carbonate saline aquifer onshore Abu Dhabi in 2023. The well was completed with fiber optic cable cemented behind the casing, allowing the monthly monitoring of CO_2 injections using DAS-VSP technology, with guaranteed repeatability on the receiver side. This paper presents the quantitative seismic interpretation aspect of the multidisciplinary study, focusing on the integration of rock physics modelling from well logs with repeated Zero-Offset VSP (ZOVSP) measurements. Rock physics analysis and fluid substitution modeling are performed to assess CO_2 detectability using elastic logs. P-wave velocity, S-wave velocity and density logs are modeled for different fluid content scenarios, starting from the in situ case with high salinity brine. A 10+% P-wave velocity decrease for 30% supercritical state CO_2 saturation is estimated, giving high confidence in detecting CO_2 injections using DAS-VSP technology. Well-to-seismic calibration is carried out using the ZOVSP baseline. An excellent match is obtained after applying up-down deconvolution, encouraging further quantitative seismic interpretation investigation. A complete analysis of the comparison of the monthly recorded ZOVSP before and during CO_2 injection is performed. An increase in time delay at the top of a large reflective layer is observed gradually once CO_2 injection has started, with an associated change of amplitude. This preliminary observation is followed by a deterministic time-lapse seismic inversion analysis. Initial models used for the inversion are updated by incorporating the time difference between the base and monitor references calculated at one horizon and converted into acoustic impedance changes. The monitor references are then inverted, highlighting a decrease in acoustic impedance at the top of the perforation zone, in magnitude comparable to those predicted by the modelling. Additionally, ultrasonic rock physics experiments are being conducted on core plugs extracted from the well to further assess and quantify the CO_2 saturation changes due to CO_2 injection at in situ conditions. This study based on ZOVSP corridor stack difference analysis validates the ability of detecting acoustic impedance variations associated with CO_2 injection in a carbonate saline aquifer using DAS-VSP technology. This opens the gate to a similar quantitative analysis to be performed on the 3D DAS-VSP data to monitor spatial changes away from the well provided velocity

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models are updated accordingly during the processing stage. This is to our knowledge the first application of such a workflow for CO_2 storage monitoring in carbonates.

Introduction

While onshore Abu Dhabi's carbonate saline aquifers offer significant potential for CO_2 sequestration, they must meet certification requirements, including CO_2 plume monitoring to ensure containment and conformance. Unfortunately, case studies have shown that onshore time-lapse seismic monitoring in the region is challenging due to the stiffness of carbonate rocks and the non-repeatability of surface seismic, with NRMS values reaching up to 50% (Kirstetter et al., 2023; Mahgoub et al., 2023). In 2023, ADNOC drilled a CO_2 sequestration well in a carbonate saline aquifer as part of its goal to reduce carbon intensity by 25% by 2030 and achieve Net Zero by 2045 (Xu et al., 2024). The well was equipped with a fiber optic cable cemented behind casing, allowing for injection monitoring using DAS-VSP technology, which guarantees repeatability on the receiver side (Cambois et al., 2024; Mason et al., 2024). CO_2 injections began in November 2023.

This paper presents the quantitative seismic interpretation aspect of the multidisciplinary study, focusing on the integration of rock physics modelling from well logs with repeated Zero-Offset VSP (ZOVSP) measurements. Specifically, it examines zero-offset VSPs (ZOVSP) recorded monthly (14 ZOVSPs, from P1 to P14) and assesses the evolution of time-delay and amplitude response at specific reflectors due to CO_2 injections. Additionally, based on a modelling exercise, it investigates how similar quantitative analysis could be applied using the 3D VSP data, to monitor spatial changes away from the well provided velocity models are updated accordingly during the processing stage. Finally, it presents preliminary ultrasonic rock physics experiments conducted on core plugs extracted from the well, under varying levels of pore pressure and fluid saturation, to quantify the CO_2 saturation changes due to CO_2 injection.

Method

A comprehensive petrophysical analysis, calibrated to core samples at target level, has been conducted. This includes the estimation of porosity, permeability, and lithologic information after successfully acquiring and processing logging-while-drilling and wireline data at the well (Karim et al., 2024). The well logs, along with the synthetic seismogram and ZOVSPs, are presented in Figure 1. Subsequently, a rock physics analysis is performed on this high-porosity carbonate saline aquifer interval. This includes fluid substitution using Gassmann's equation (Gassmann, 1951) to assess the detectability of fluids (in this case supercritical CO_2 in brine) using elastic logs. P-wave velocity, S-wave velocity, and density logs are modeled for various fluid content scenarios, starting with the in situ case (high salinity brine). The results indicate an expected reduction in P-wave velocity of up to 10% for a 30% CO_2 injection in a supercritical state. These findings increased our confidence in detecting CO_2 injection effects using seismic measurements in this carbonate saline aquifer.

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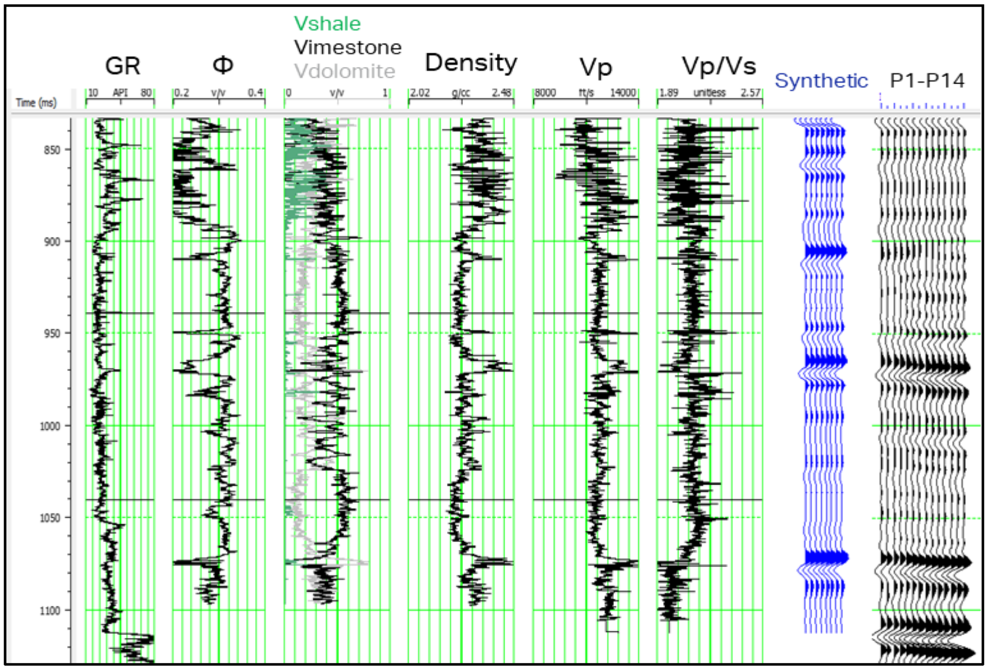

Figure 1—Well log data (Gamma Ray, porosity, lithology, density, Vp and Vp/Vs) are displayed together with a synthetic seismogram and ZOVSP corridor stacks (P1 to P14). Top and bottom perforation depths are indicated by the horizontal black lines (respectively 940ms and 1040ms).

The next phase involves well-to-seismic calibration, using the ZOVSP baseline (P1) corridor stack as a reference. An excellent match is achieved between P1 and the synthetic seismogram, with an 84% correlation at the reservoir level and a stable deterministic wavelet. This correlation further increases to 93% after applying up-down deconvolution to the ZOVSP data, which eliminates the remaining multiple energy (Figure 2).

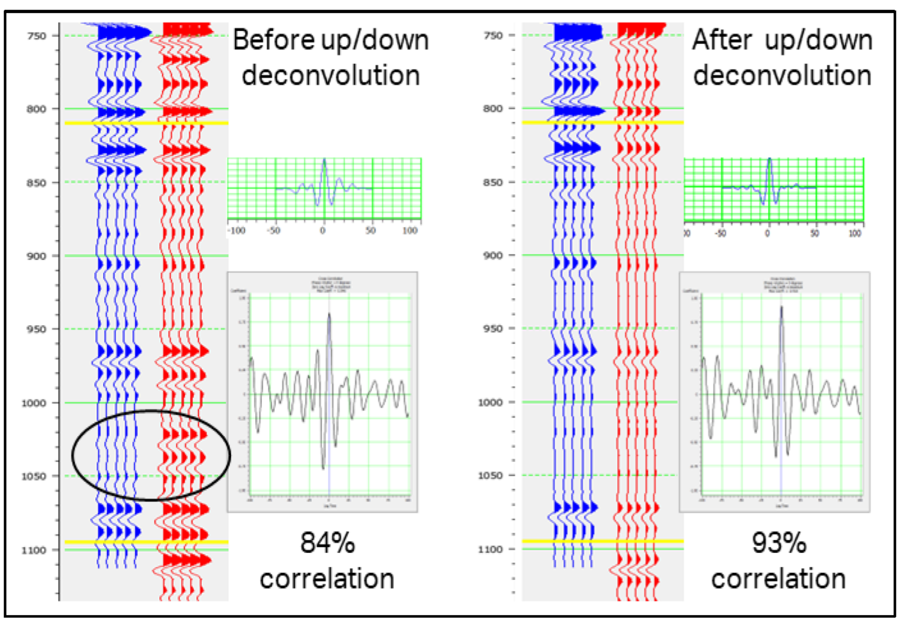

Figure 2—Zero offset corridor stack trace P1 (red) compared to the synthetic seismogram (blue) before and after up/down deconvolution. The remaining multiples before up/down deconvolution are highlighted by the black ellipse

A comparison of the monthly recorded ZOVSP data (P1 to P14) is then conducted. CO_2 injection began after P3, and an increase in time delay at the top of a large reflective layer is clearly observed after P4,

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along with an associated change in amplitude. An initial estimate of the velocity decrease over time at the injection zone is estimated by flattening the main reflector (965ms) at P1 arrival-time. An analysis of the amplitude evolution over time at several levels is also performed, showing a response consistent with CO_2 injection at the reservoir level, with a decrease in amplitude over time at the top of the effective injection zone and an increase at the bottom of the injection zone (Figure 3).

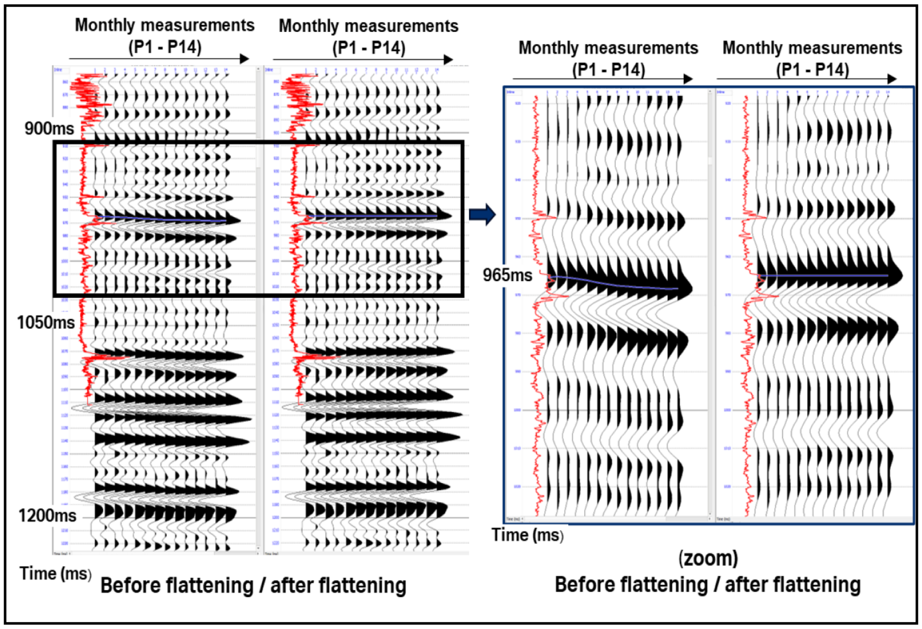

Figure 3—ZOVSP traces from P1 to P14 before and after flattening at the 965ms blue event. The acoustic impedance log is displayed in red. It is observed that deeper events below 1050ms are also aligned after the flattening process indicating no change of velocity in the deeper part of the injection zone.

Encouraged by the high correlation between the well synthetic seismogram and the P1 ZOVSP corridor stack, we then conduct a time-lapse seismic inversion analysis. The initial acoustic impedance model for each monitor survey is computed by incorporating the time difference between the base and monitor references calculated at one horizon and converting it into P-wave velocity and acoustic impedance changes. The monitor reference is then inverted, and results are presented below.

Results

Figure 4 illustrates the changes in acoustic impedance from P1 to P14, showing a noticeable decrease at the top of the perforation zone, just above a low permeability layer. This decrease is consistent with the predictions from the modeling, which indicated an average velocity reduction of 10%. The analysis suggests that CO_2 injection occurs primarily in the upper part of the perforation zone and slightly above it, rather than throughout the entire perforation zone as initially anticipated. This finding led us to reassess the initial understanding of the CO_2 plume's extent from the well and prompted a review of the placement of shallow soil CO_2 onsite tracer sensors, moving them further from the borehole to meet HSE and assurance certification requirements.

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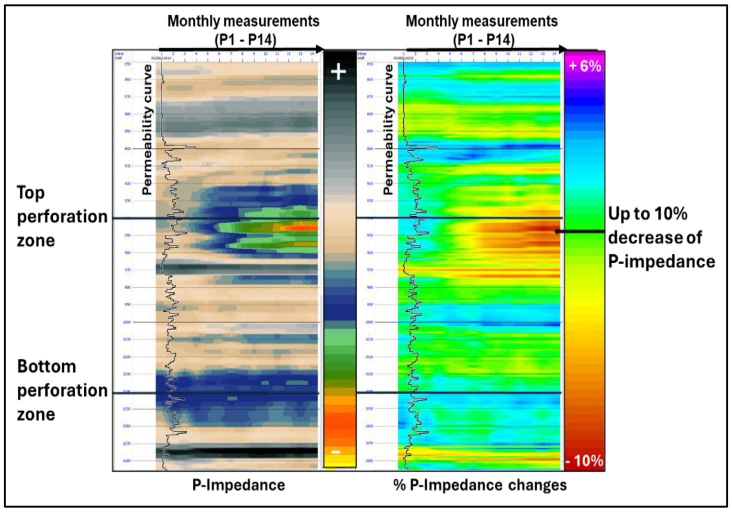

Figure 4—Acoustic impedance section derived from time-lapse seismic inversion of ZOVSP traces (P1 to P14). Traces have been recorded monthly. The permeability curve is plotted in black. CO_2 injections started between P3 and P4. This highlights that CO_2 injection is effectively occurring at the top part of the perforation zone only, mainly above the low permeability layer, rather than on the entire length of the perforation zone.

Additionally, 3D DAS-VSP surveys were acquired and processed before and after the start of CO_2 injections to monitor the spatial extension of the CO_2 plume. The main processing steps applied to the 3D DAS-VSP data included deblending, static corrections, surface-consistent deconvolution, and depth migration. Preliminary analysis of the comparison of the 3D DAS-VSP data revealed a time delay increase around the well within the injection zone with time commensurate with what was observed in the ZOVSP (Figure 5). This highlights that 3D DAS-VSP technology is able to monitor the spatial extension of the CO_2 plume in this saline carbonate aquifer. As an additional step to the qualitative estimation of the time-shift due to CO_2 injections presented in Figure 5, subsequent analysis is necessary to estimate quantitatively the P-impedance changes as previously done in the case of the ZOVSP.

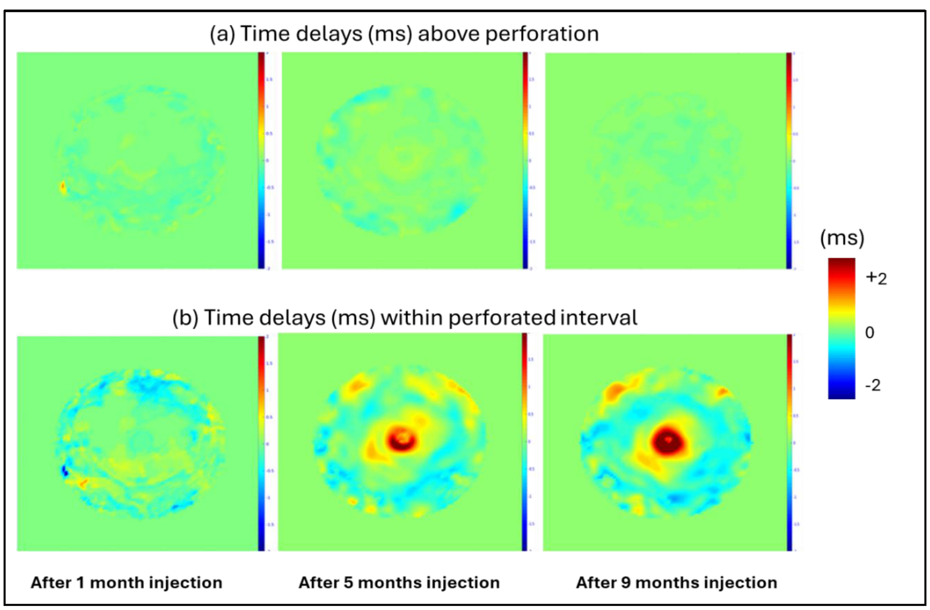

Figure 5—Maps of time delays (ms) around the well above the injection zone (a) and inside the injection (b) between 3D DAS-VSP base and monitors. The first, second and third repeat surveys were acquired respectively one, five and nine months after the start of CO_2 injection. The contour of the CO_2 plume is clearly visible after five months of injection within the injection zone.

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Walkaway-VSP modelling

To further investigate the ability to delineate the CO_2 plume migration in 3D using DAS-VSP technology, a walkaway-VSP modelling was carried out to assess the impact of the velocity anomaly resulting from the CO_2 injection on VSP imaging and its time-lapse response. Starting from a 1D model extrapolated from the well, two 1.5D velocity models were built: one with no CO_2 anomaly, one with a 10% velocity decrease around the well due to the presence of CO_2 (Figure 6). Walkaway-VSP data were modelled for both cases (before and after CO_2 injection, respectively base and monitor) using a finite difference modeling algorithm (133 Shots, 66 on each side of the well, with a 15m shot spacing, 990m max offset).

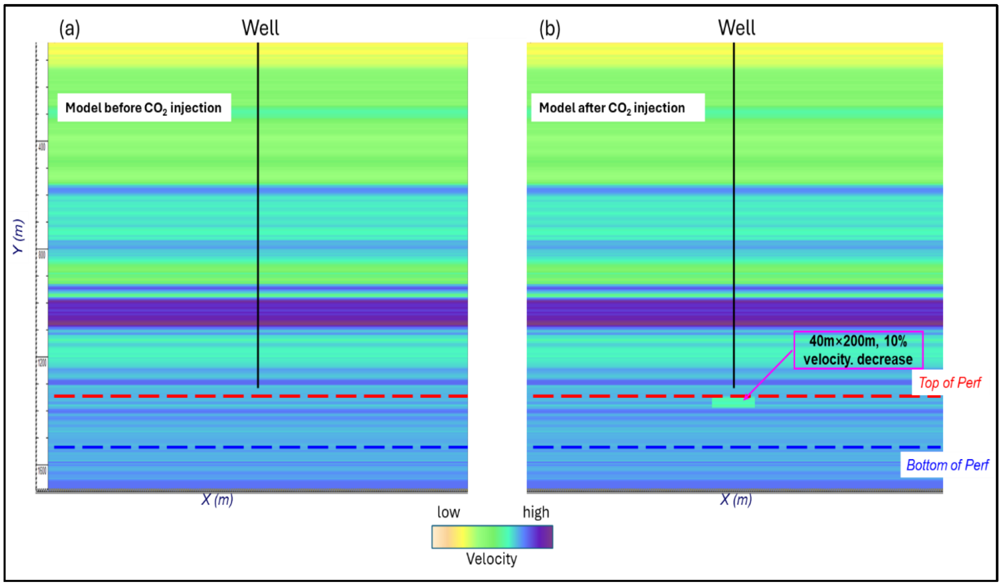

Figure 6—Velocity model without (a) and with (b) velocity anomaly due to presence of CO_2.

Depth-migrated images were generated using an RTM (Reverse Time Migration) algorithm, before and after CO_2 injection, and the time-lapse image was derived by trace-by-trace difference in the time domain after depth to time conversion. Two cases were considered: the first case (a) using the initial velocity model for both base and monitor RTM, and a second case (b) using the correct velocity model (after CO_2 injection) for the monitor RTM, whereas the initial velocity model was used for the base RTM. In both cases (a) and (b) for both base and monitor modeling, the depth to time conversion was carried out using the initial velocity model (before CO_2 injection). Time-lapse sections for both cases are presented in Figure 7. It shows an accurate time-lapse signal corresponding to the anomaly limits for case (b), for which the correct velocity model was used for the RTM, whereas the time-lapse signal for case (a) is contaminated by leakage due to the fact the velocity model was not updated accordingly. It clearly indicates the necessity to derive the updated velocity model due to CO_2 injection to compute a correct quantitative time-lapse volume.

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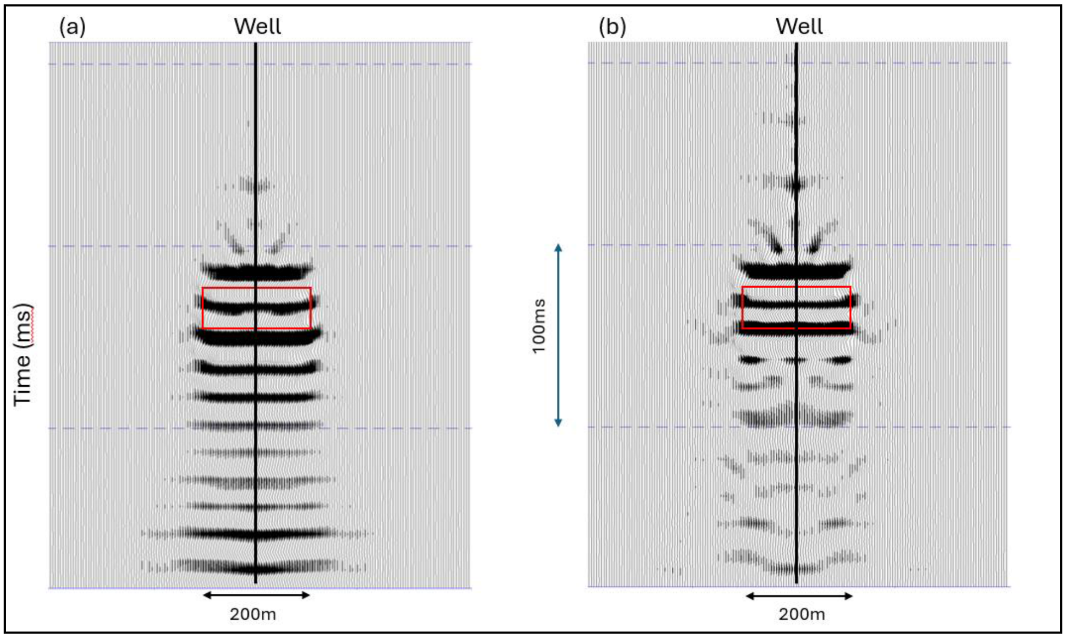

Figure 7—RTM time-lapse difference in time. (a) Identical velocity model for both base and monitor RTM (no velocity update). (b) Updated velocity model used for monitor RTM. The red box indicates the limits of the velocity anomaly due to CO_2 injection.

Rock Physics Laboratory Experiments

ZOVSP inversion allowed us to quantify the CO_2 injection in terms of acoustic impedance changes, as demonstrated previously. To further advance the quantification of the CO_2 response, a program of laboratory rock physics has been designed with the objective of relating these changes to CO_2 saturation and pressure changes due to CO_2 injection. Core plugs from the well in the injection zone of the aquifer formation were selected and sent to ADNOC Research and Innovation Center (ADRIC) for ultrasonic rock physics measurements. This included a total of 7 core plugs, with porosity and permeability ranging from 25 to 35% and 40 to 100mD. After careful sample preparation, including cleaning of the plugs, ultrasonic rock physics measurements were performed. The Autolab-1000 acoustics rig (manufactured by New England Research, Vermont, USA) was used to measure P- and S-wave velocities for different pressure and fluid saturation conditions on 2,5" by 1,5" diameter cylindrical core samples, using transducer core holders operating at frequency of 500KHz to 1MHz.

The program includes the following three stages:

- 1). Ultrasonic measurements on dry core plugs, varying the confining pressure up to values representative of those in situ (see Janssen et al., (2025) showing measurements during two cycles of loading and unloading at dry conditions);
- 2). Ultrasonic measurements on 100% brine-saturated core plugs (using synthetic brine), varying both confining pressure and effective stress up to values representative of those in situ;
- 3). Ultrasonic measurements on core plugs with supercritical CO_2 injection into 100% brine-saturated samples, at confining pressure and effective stress values representative of those in situ, varying gradually the saturation of the supercritical CO_2.

The main benefit from these experiments is that both pressure and fluid saturation are fully controlled when measuring P- and S-wave velocities, allowing trends to be extracted as a function of effective stress and fluid saturation when varying confining pressure, pore pressure and fluid saturation. Examples of such measurements involving supercritical CO_2 injections are numerous in the literature on sandstones (e.g.,

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Lebedev et al., 2013), however no example of such a program on carbonates could be found. Preliminary results of phases 1) and 2) are presented in Figures 8 and 9 for one core plug representative of the study.

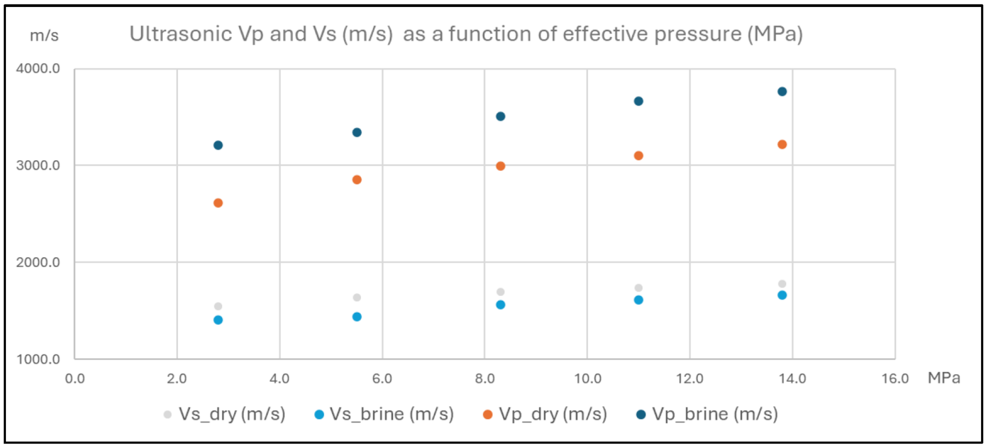

Figure 8—Experimental ultrasonic Vp (m/s) and Vs (m/s) measurements as a function of effective pressure for brine saturated and dry conditions. Brine saturation leads to an increase in Vp and a decrease in Vs. Both Vp and Vs increase with effective pressure, whether in dry or brine saturated conditions.

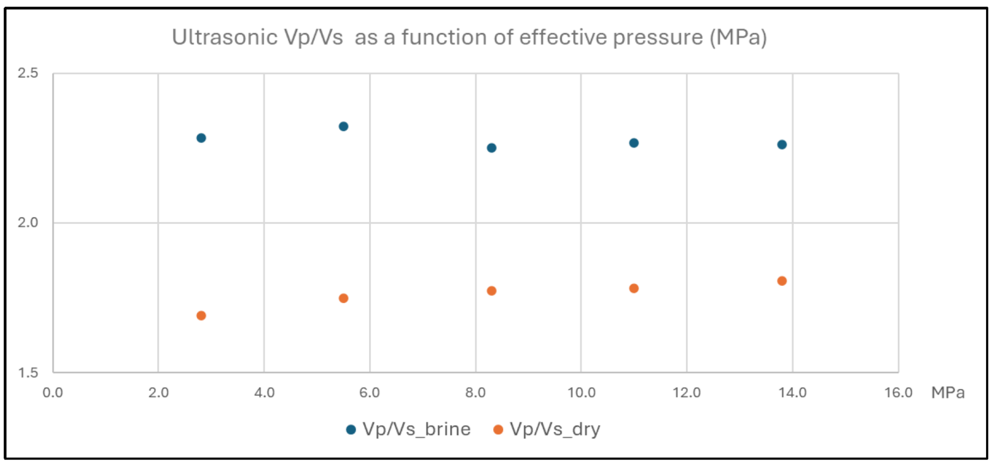

Figure 9—Experimental ultrasonic Vp/Vs as a function of effective pressure for brine saturated and dry conditions. Higher values of Vp/Vs are observed at brine saturated conditions compared to those measured at dry conditions, whereas constant values are observed with increasing confining pressure.

Maximum confining pressure and effective stress applied on the core plugs do not exceed those in situ, allowing the preservation of core plugs integrity for additional testing, and ensuring their stress and strain conditions remain in the elastic domain. As expected, results highlight an increase in P-wave and S-wave velocities as a function of increasing confining pressure. Also, an increase in P-wave velocity is observed in brine saturation compared to dry conditions, with a Vp/Vs ratio increase as predicted by modelling. Comparison of these measurements made on the 100% brine-saturated core plugs with those predicted by theoretical Gassmann fluid substitution using velocities measured on the core plugs under dry conditions is ongoing. The next step will consist of injecting supercritical CO_2 into the brine-saturated core plugs while measuring the ultrasonic P- and S-wave velocity for varying CO_2 saturation levels, simulating the experiments occurring in the field.

An additional phase is also considered with the objective of assessing the possible long-term impact of CO_2/brine rock interactions on mineral dissolution and/or carbon mineralization on the dry frame of the core plugs, as reported by Adam et al. (2015) and Vanorio (2015). This phase will involve setting up an experiment to age carbonate core plugs in supercritical CO_2/brine for various time periods. Ultrasonic

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measurements will be performed at regular intervals during the aging process and will be compared with similar measurements carried out on core plugs without any aging.

Conclusions

This study demonstrates the potential for monitoring CO_2 injection in a carbonate saline aquifer using DAS-VSP technology. By using seismic quantitative interpretation techniques (including time-lapse inversion) on ZOVSP data acquired before and during CO_2 injections, estimation of velocity and acoustic impedance due to CO_2 injection was made possible with changes of an order of magnitude similar to those theoretically predicted. Additional results using 3D DAS-VSP data recorded every four months provide qualitative maps of the extension of the CO_2 plume. This paves the way for further quantitative analyses to be conducted on the 3D VSP data. 2D walkaway-VSP modelling highlights that additional processing including velocity model updating with CO_2 injection is needed to quantify the time-lapse effects using 3D DAS-VSP, and to extract quantitative properties in 3D thus allowing monitoring of the plume's spatial distribution and its elastic property variations with injection. State-of-the-art 4D FWI technology is believed to be the desired technology for such a purpose. Finally, as an attempt to further enhance the quantitative aspect linking the acoustic impedance changes to CO_2 saturation and pressure variations, a comprehensive program of advanced ultrasonic rock physics experiments on core plugs is progressing. Measurements carried out on both dry and brine-saturated samples have been performed with the next step being to perform the same measurements when injecting supercritical CO_2 into the brine-saturated core plugs, replicating the field experiment. This will allow the derivation of empirical trends of P- and S- wave velocities as a function of CO_2 saturation to be integrated with the quantitative results from ZOVSP and 3D DAS-VSP to help constrain the reservoir simulation dynamic model.

Acknowledgements

The authors thank ADNOC management for permission to publish this paper, BGP for both modeling and processing the VSP data and ADRIC for carrying out the laboratory experiments.

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